A spectral method for multi-view subspace learning using the product of projections

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Summary

Multi-view data provides complementary information on the same set of observations, with multi-omics and multimodal sensor data being common examples. Analyzing such data typically requires distinguishing between shared (joint) and unique (individual) signal subspaces from noisy, high-dimensional measurements. Despite many proposed methods, the conditions for reliably identifying joint and individual subspaces remain unclear. We rigorously quantify these conditions, which depend on the ratio of the signal rank to the ambient dimension, principal angles between true subspaces, and noise levels. Our approach characterizes how spectrum perturbations of the product of projection matrices, derived from each view's estimated subspaces, affect subspace separation. Using these insights, we provide an easy-to-use and scalable estimation algorithm. In particular, we employ rotational bootstrap and random matrix theory to partition the observed spectrum into joint, individual, and noise subspaces. Diagnostic plots visualize this partitioning, providing practical and interpretable insights into the estimation performance. In simulations, our method estimates joint and individual subspaces more accurately than existing approaches. Applications to multi-omics data from colorectal cancer patients and nutrigenomic study of mice demonstrate improved performance in downstream predictive tasks.

Some key words: data integration, random matrix theory, rotational bootstrap, spectral analysis, subspace estimation

1. Introduction

1.1. *Background*

Multi-view data provides complementary information on the same set of observations, with multi-omics and multimodal sensor data being common examples. Since each data source or view is potentially high-dimensional, it is of interest to identify a low-dimensional representation of the signal and further distinguish whether the signal is shared (joint) across views or view-specific (individual). For example, in § 6.1, we consider colorectal cancer data with RNAseq and miRNA views. The goal is to identify joint signals from these two views and validate whether these signals predict cancer subtypes (Guinney et al., 2015).

Canonical correlation analysis (Hotelling, 1936) is a time-tested approach for finding associated signals across views, but it cannot distinguish individual signals. To address this limitation, many methods have been developed to explicitly model joint and individual signals (Van Deun et al., 2009; Lock et al., 2013; Zhou et al., 2015; Yang & Michailidis, 2016; Feng et al., 2018; Shu et al., 2019; Gaynanova & Li, 2019; Park & Lock, 2020; Murden et al., 2022; Prothero et al., 2024; Xiao & Xiao, 2024).

Despite significant methodological advances, a crucial theoretical question remains: when can joint and individual signals be reliably identified from multi-view data? To answer this question, we start with the two-view case and adopt the definition of joint and individual signal subspaces following Lock et al. (2013); Zhou et al. (2015); Feng et al. (2018). In this framework, the joint signals correspond to the same subspace and are orthogonal to individual signals, whereas the individual signals do not intersect (albeit may not be orthogonal). Conceptually, it is clear that (1) separation of joint and individual signals becomes difficult when the angle between individual subspaces gets small; (2) the total rank of the signal must be small to avoid noisy directions being mistaken for the joint due to random overlaps. However, the precise quantification of these concepts has been lacking.

1.2. *Our contributions*

Our main contribution is an explicit characterization of conditions under which the joint and individual signals are identifiable, which depend on the signal rank, noise level, and principal angles between individual subspaces. Our primary insight is that the spectrum perturbations of the *product of projection matrices*, derived from each view's estimated subspaces, affect subspace separation. Figure 1 provides a visual illustration of this theoretical result under varying subspace alignment and noise levels. The observed spectrum of the product of projections, which corresponds to the cosine of principal angles between the two subspaces, exhibits a clustering structure that corresponds exactly to joint and individual subspaces. Specifically, the largest singular values close to 1 correspond to joint subspace, the middle singular values correspond to non-orthogonal individual subspaces, and the smallest singular values correspond to orthogonal individual and noise directions. The bars indicate the observed spectrum, whereas the colored areas highlight the theoretically derived predictions on the spectrum perturbation in alignment with observed clustering. A lack of interval overlap supports the idea that joint subspaces are distinguishable from individual subspaces and noise.

Motivated by new theoretical insights on the perturbation of the spectrum of the product of projections, we propose an estimation approach that quantifies these perturbations in practice with an easy-to-use and scalable algorithm. First, we develop a novel rotational bootstrap technique to bound the perturbations of the spectrum corresponding to joint and non-orthogonal individual subspaces (top two clusters). Second, we utilize results from random matrix theory to quantify the maximal alignment possible by chance, providing a bound on the bottom cluster (orthogonal individual and noisy subspace directions). We use these two bounds to obtain estimates for the joint and individual subspaces. Further, we generate diagnostic plots that visualize cluster partitioning with the two bounds overlaid (like the one in Figure 1), providing practical and interpretable insights regarding the identifiability of joint and individual subspaces.

In simulations, our method is competitive compared to those of Lock et al. (2013); Feng et al. (2018); Gaynanova & Li (2019); Shu et al. (2019); Park & Lock (2020), particularly in terms of

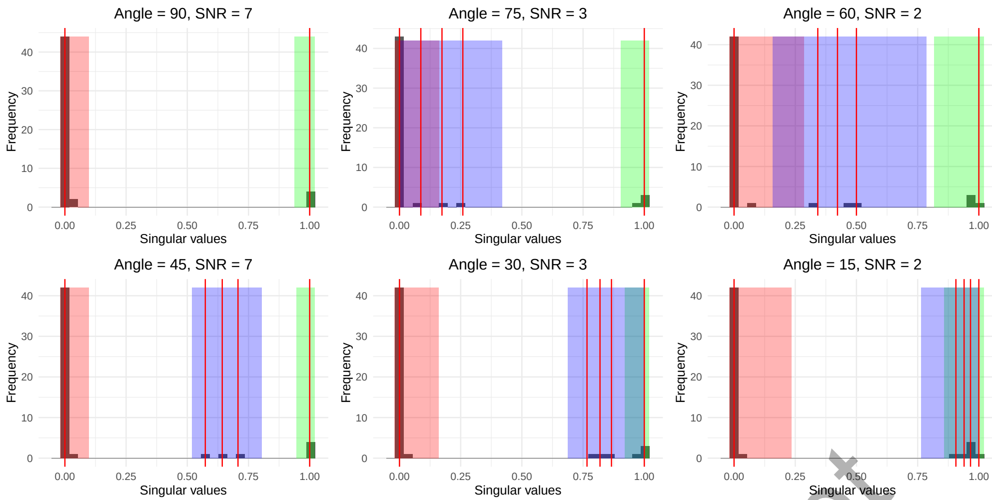

Fig. 1: The alignment between the observed spectrum of the product of projections (grey histogram) and theoretical predictions from Theorem 1 (highlighted blocks) under varying angles between individual subspaces and signal-to-noise ratios (SNR). The theoretical intervals correspond to the joint (green), non-orthogonal individual (blue), and orthogonal individual and noise (red) components. The vertical red lines show the singular values in the noiseless setting.

true and false discovery rates in estimating joint and individual subspaces. Additionally, we test our approach on colorectal cancer data and nutrigenomic data. The colorectal cancer data includes tumor samples with RNASeq and miRNA views, where we aim to identify joint components. We assess the quality of these components through their predictive power for cancer subtype classification. The nutrigenomic data consists of mice with distinct genotypes and diets, with RNASeq and liver lipid content as the two views. Here, we seek to identify joint and individual components and link them to genotype and diet labels, respectively. In both datasets, our approach achieves the best balance between parsimony and predictive accuracy.

1.3. *Related works*

Lock et al. (2013); Zhou et al. (2015); Gaynanova & Li (2019); Park & Lock (2020); Yi et al. (2023); Prothero et al. (2024); Xiao & Xiao (2024) consider optimization-based approaches for estimating joint and individual subspaces. While effective, these methods can be computationally slow, particularly with high-dimensional data, and often rely on ad-hoc techniques for rank selection (such as permutation or bi-cross-validation). Xiao & Xiao (2024) is an exception in that it provides a rigorous theoretical analysis of estimation consistency for its optimization-based approach. However, its methodological framework is based on sparse principal component analysis and on estimating eigenvectors corresponding to features. In contrast, our work focuses on joint and individual subspaces, corresponding to samples, based on spectral methods. Feng et al. (2018), Shu et al. (2019) and Chen et al. (2022) are spectral methods, like ours, and utilize singular value decomposition to derive joint and individual subspaces. Shu et al. (2019) extends canonical correlation analysis to account for individual components. However, the individual components are restricted to be orthogonal, leading to a different model from ours. Feng et al. (2018) uses the spectrum of the average of projection matrices, which contains information about the principle angles between the two subspace views, to estimate joint and individual components. The method allows for non-orthogonal individual signals and is most similar to ours; however, it has certain limitations in comparison, as we discuss below. Chen et al. (2022) consider a different data integration setting where data sets are linked by features rather than matched by samples, using a 2×2 factorial design. Similar to Feng et al. (2018), their method relies on the spectrum of the average of projection matrices for estimation.

Conceptually, Feng et al. (2018) and Chen et al. (2022) consider the average of projection matrices, while our method is the first to focus on the product of projection matrices. A key disadvantage of using the average of projection matrices is that its spectrum exhibits a provably smaller gap (by a factor of less than one-half in noiseless settings) between individual and joint components than the product. This results in a substantially worse clustering of singular values in practice, especially in moderate to low signal-to-noise regimes (see e.g, Figure 1 in Appendix A.6). The lack of clustering in the spectrum considered by Feng et al. (2018) makes separating the joint directions from the remaining signals more challenging. See § 5 for more numerical comparison and Appendix A.6 for more technical details.

2. MODEL

2.1. *Notation*

For $a, b \in \mathbb{R}$, let $a \wedge b = \min(a, b)$, $a \vee b = \max(a, b)$. For a matrix $A \in \mathbb{R}^{n\times p}$, write the compact singular value decomposition (SVD) as $A = U\Sigma V^T$, where $\Sigma = \mathrm{diag}(\sigma_1(A), \sigma_2(A), \dots)$ with the singular values $\sigma_1(A) \geq \sigma_2(A) \geq \cdots > 0$ in descending order. We denote $\sigma_{\min}(A) = \sigma_{\mathrm{rank}(A)}(A)$, $\sigma_{\max}(A) = \sigma_1(A)$ as the smallest and largest non-trivial singular values. We denote $\sigma(A)$ to be the collection $\{\sigma_1(A), \sigma_2(A), \dots, \sigma_{\mathrm{rank}(A)}(A)\}$. We use $\|A\|_2 = \sigma_1(A)$ to denote the spectral norm, $\|A\|_F$ to denote the Frobenius norm and $\mathrm{col}(A)$ to denote the column space of A. We write the subspaces in calligraphic, e.g., we say $\mathcal{A} \subset \mathbb{R}^n$ to denote a subspace of $\mathbb{R}^n$. For the projection onto a subspace $\mathcal{A}$, we write $P_{\mathcal{A}} \in \mathbb{R}^{n\times n}$. For the projection onto the column space of A, we use $P_{\mathrm{col}(A)}$, which can be defined as $P_A = UU^\top$. We denote the direct sum of two subspaces $\mathcal{A}$ and $\mathcal{B}$ by $\mathcal{A} \oplus \mathcal{B}$. We adopt the convention to denote the submatrix: $A_{[a:b,c:d]}$ represents the a-to-b-th row, c-to-d-th column of matrix A; we also use $A_{[a:b,:]}$ and $A_{[:,c:d]}$ to represent a-to-b-th full rows of A and c-to-d-th full columns of A, respectively. We use the Bachman-Landau symbols O to describe the limiting behavior of a function (in terms of sample size and number of features). Further, we use $z \asymp 1$ to express $z = O(1)$ and $1/z = O(1)$. Finally, we use $z_1 \gtrsim z_2$ to mean that $z_1 \geq cz_2$ for some constant $c > 0$.

2.2. *Setup*

Let $Y_1 \in \mathbb{R}^{n\times p_1}$, $Y_2 \in \mathbb{R}^{n\times p_2}$ be observed data matrices corresponding to two views. The rows of both matrices represent common n subjects, and the columns correspond to the different feature sets collected for each view. For example, in the COAD data (see § 6.1), the first view corresponds to the RNASeq features and the second to the miRNA features. In the nutrigenomic study (see § 6.2), the first view corresponds to the gene expressions and the second view corresponds to the lipid content of the liver. We assume that Y_1, Y_2 are noisy observations of true X_1, X_2, and consider additive structured signal plus noise decomposition:

$$Y_k = X_k + Z_k = J_k + I_k + Z_k, \tag{1}$$

where the signal X_k is decomposed into joint part across views (J_k) and the view-specific individual part (I_k), and Z_k is the noise matrix with elements identically and independently distributed according to some distribution.

Model (1) is adopted by many existing methods for multi-view data integration (Lock et al., 2013; Feng et al., 2018; Gaynanova & Li, 2019; Shu et al., 2019; Park & Lock, 2020; Yi et al., 2023; Prothero et al., 2024) with some variations in underlying definitions of joint and individual signals. Here we adopt the original definition as in Lock et al. (2013). To facilitate exposition, consider singular value decomposition representation of each part of the signal:

$$X_k = J_k + I_k = U_{\mathcal{J}_k}\Sigma_{\mathcal{J}_k}V_{\mathcal{J}_k}^\top + U_{\mathcal{I}_k}\Sigma_{\mathcal{I}_k}V_{\mathcal{I}_k}^\top. \tag{2}$$

We refer to $U_{\mathcal{J}_k}$ as joint directions and to $U_{\mathcal{I}_k}$ as individual directions. Joint directions are joint due to the equality of corresponding subspaces, that is they correspond to the joint subspace $\mathcal{J} := \mathrm{col}(U_{\mathcal{J}_1}) = \mathrm{col}(U_{\mathcal{J}_2})$. Thus, joint $U_{\mathcal{J}_1}$ can be reconstructed from joint $U_{\mathcal{J}_2}$ by a rotation by some orthogonal matrix. In contrast, individual directions have non-intersecting subspaces $\mathcal{I}_k := \mathrm{col}(U_{\mathcal{I}_k})$, that is $\mathcal{I}_1 \cap \mathcal{I}_2 = \{\mathbf{0}\}$, which emphasizes that they are indeed specific to each view. For identifiability, joint and individual subspaces are assumed to be orthogonal, that is $\mathcal{J} \perp \mathcal{I}_k$.

However, the individual $\mathcal{I}_1$ and $\mathcal{I}_2$ are not necessarily orthogonal and can be further decomposed into orthogonal and non-orthogonal parts, leading to a three-part decomposition (joint, non-orthogonal individual, orthogonal individual). We further provide explicit characterization and identifiability conditions of this three-part decomposition of signal column spaces, which is based on Lemma 1 of Feng et al. (2018) and Proposition 4 of Gaynanova & Li (2019).

Lemma 1 (Existence and Identifiability). *Given a set of subspaces* $\{\mathrm{col}(X_1), \mathrm{col}(X_2)\}$*, there is a unique set of subspaces* $\{\mathcal{J}, \mathcal{N}_1, \mathcal{N}_2, \mathcal{O}_1, \mathcal{O}_2\}$ *such that:*

1. $\mathrm{col}(X_k) = \mathcal{J} \oplus \mathcal{N}_k \oplus \mathcal{O}_k$ *with individual* $\mathcal{I}_k := \mathcal{N}_k \oplus \mathcal{O}_k$*;*
2. $\mathcal{J} \perp \mathcal{I}_k$ *for all* $k \in \{1,2\}$*,* $\mathcal{N}_k \perp \mathcal{O}_l$ *for all* $k, l \in \{1,2\}$*;*
3. $\mathcal{O}_1 \perp \mathcal{O}_2$*;*
4. $\bigcap \mathcal{N}_k = \{\mathbf{0}\}$ *and maximal principle angle between* $\mathcal{N}_1$ *and* $\mathcal{N}_2$ *is* $< \pi/2$*.*

Here, the individual subspace is given by $\mathcal{I}_k = \mathcal{N}_k \oplus \mathcal{O}_k$, with $\mathcal{N}_k$ corresponding to non-orthogonal part and $\mathcal{O}_k$ to orthogonal part. The three-part decomposition can be viewed from the perspective of ordering the principal angles between subspaces $\mathrm{col}(X_1)$ and $\mathrm{col}(X_2)$ (§ 3.2). Specifically, it can be viewed as aligning the basis of $\mathrm{col}(X_1)$ and $\mathrm{col}(X_1)$ into pairs, from perfectly aligned pairs (zero angles, joint subspace $\mathcal{J}$) to somewhat aligned pairs (angles strictly between 0 and $\pi/2$, non-orthogonal $\mathcal{N}_1$ and $\mathcal{N}_2$) to orthogonal basis that can not be paired ($\pi/2$ angles, orthogonal $\mathcal{O}_1$ and $\mathcal{O}_2$).

Our main goal is to distinguish joint subspace from individual based on the noisy observed Y_k. Conceptually, it is clear that (1) separation of joint and individual signals becomes difficult when the angle between individual subspaces gets small, i.e. the angle between $\mathcal{N}_1$ and $\mathcal{N}_2$; (2) the total rank of the signal must be small to avoid noisy directions being mistaken for the joint signal due to random overlaps. However, the precise quantification of these concepts has been lacking. Next, we illustrate how the spectrum of the product of projection matrices allows us to distinguish joint subspaces from the aligned individual ones and provide theoretical quantification of the spectrum perturbation as a result of noise. We then use these insights to develop an effective and interpretable estimation approach for joint and individual subspaces.

3. A New Perspective: Product of Projections Decomposition

3.1. *Overview*

Our theoretical insights, which subsequently motivate the proposed estimation approach, rely crucially on the product of projection matrices associated with each view's subspace. The spectrum of this product corresponds to the cosines of principal angles between the pair of subspaces (Bjoerck & Golub, 1971). In noiseless cases, this spectrum can be clustered to perfectly separate joint and individual subspaces (§ 3.2). In the presence of noise, this spectrum is perturbed, and in § 3.3, we quantify these perturbations with respect to each cluster, leading to explicit conditions for correct estimation of joint rank and quantification of corresponding subspace estimation error. Throughout, we focus on the case of two views, with Appendix A.5 describing the extension to the larger number of views.

3.2. *Product of projections in the noiseless setting*

Consider the noiseless case with $Z_k = \mathbf{0}_{n\times p_k}$ so that $Y_k = X_k$. Let $X_k = U_k\Sigma_k V_k^\top$ be the compact singular value decomposition of signal X_k. Using Lemma 1, we have:

$$P_{\mathrm{col}(X_k)} = U_k U_k^\top = P_{\mathcal{J}} + P_{\mathcal{I}_k} \quad \Rightarrow \quad P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)} = P_{\mathcal{J}} + P_{\mathcal{I}_1}P_{\mathcal{I}_2}.$$

Consider the second term $P_{\mathcal{I}_1}P_{\mathcal{I}_2}$. Let $U_{\mathcal{J}} \in \mathbb{R}^{n\times \dim(\mathcal{J})}$ be the matrix whose columns form orthonormal basis of $\mathcal{J}$, similarly let $U_{\mathcal{N}_1}, U_{\mathcal{N}_2}, U_{\mathcal{O}_1}, U_{\mathcal{O}_2}$ be the matrices of the orthonormal bases for $\mathcal{N}_1, \mathcal{N}_2, \mathcal{O}_1, \mathcal{O}_2$. Then,

$$P_{\mathcal{I}_1}P_{\mathcal{I}_2} = (P_{\mathcal{N}_1} + P_{\mathcal{O}_1})(P_{\mathcal{N}_2} + P_{\mathcal{O}_2}) = P_{\mathcal{N}_1}P_{\mathcal{N}_2} = U_{\mathcal{N}_1}U_{\mathcal{N}_1}^\top U_{\mathcal{N}_2}U_{\mathcal{N}_2}^\top = \left(U_{\mathcal{N}_1}H\right)\Sigma\left(K^\top U_{\mathcal{N}_2}^\top\right).$$

Here, $H\Sigma K^T$ is the singular value decomposition of $U_{\mathcal{N}_1}^\top U_{\mathcal{N}_2}$; diagonal entries of Σ equal cosines of the non-zero principal angles between $\mathcal{I}_1$ and $\mathcal{I}_2$. Since $\mathcal{I}_1 \cap \mathcal{I}_2 = \{\mathbf{0}\}$, $\sigma_{\max}(\Sigma) < 1$. Thus,

$$P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)} = \begin{bmatrix} U_{\mathcal{J}} & U_{\mathcal{N}_1}H \end{bmatrix} \begin{bmatrix} I_{\dim\mathcal{J}} & 0 \\ 0 & \Sigma \end{bmatrix} \begin{bmatrix} U_{\mathcal{J}} & U_{\mathcal{N}_2}K \end{bmatrix}^\top,$$

where the matrices $\begin{bmatrix} U_{\mathcal{J}} & U_{\mathcal{N}_1}H \end{bmatrix}$ and $\begin{bmatrix} U_{\mathcal{J}} & U_{\mathcal{N}_2}K \end{bmatrix}$ have orthonormal columns. Thus, the spectrum of $P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)}$ clusters into two components: a cluster with singular values equal to one and a cluster with singular values in the range $[\sigma_{\min}(\Sigma), \sigma_{\max}(\Sigma)]$. Let r_{clust1} be the number of singular values in the first cluster (corresponding to singular values one). Then, the joint and individual subspaces can be exactly identified via:

$$\begin{aligned} \mathcal{J} &= \text{ span of the first } r_{\mathrm{clust1}} \text{ left (or right) singular vectors of } P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)}, \\ \mathcal{I}_k &= \text{ span of the first } \operatorname{rank}(X_k) - r_{\mathrm{clust1}} \text{ left (or right) singular vectors of } P_{\mathrm{col}(X_k)}(I - P_{\mathcal{J}}). \end{aligned}$$

Thus, in the noiseless setting, the spectrum of the product of projections $P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)}$ (equivalently, the collection of cosines of principle angles between $\mathrm{col}(X_1)$ and $\mathrm{col}(X_2)$) can be divided into two sets. The first set contains values equal to one (equivalently, principle angles 0), whose corresponding singular vectors form a basis for the joint subspace. The second set contains values strictly less than one (equivalently, principle angles strictly greater than 0), reflecting the degree of alignment between the individual subspaces $\mathcal{I}_1$ and $\mathcal{I}_2$. The gap between these two sets, formally defined as one minus the cosine of the smallest principle angle between $\mathcal{I}_1$ and $\mathcal{I}_2$, narrows as the alignment between individual subspaces increases.

3.3. *Deterministic analysis of the product of projection matrices in a noisy setting*

In practice, we do not have direct access to the subspace $\mathrm{col}(X_k)$ because the signal X_k is unknown. Instead, we have to estimate $\widehat{X}_k$ from noisy observations, with a truncated singular value decomposition of Y_k being a common approach (Feng et al., 2018; Shu et al., 2019). However, the spectrum of the estimated product $P_{\mathrm{col}(\widehat{X}_1)}P_{\mathrm{col}(\widehat{X}_2)}$ differs from the spectrum of the true product $P_{\mathrm{col}(X_1)}P_{\mathrm{col}(X_2)}$ in several key ways. First, even when a joint signal is present, the largest singular values in the estimated product are generally below one and decrease further as the noise level or the dimension of the individual views increases. Second, in the estimated product, the gap between the singular values corresponding to the joint subspaces and those corresponding to the aligned individual subspaces is typically smaller than in the true product. This gap also decreases as noise or dimensionality increases, making it harder to separate joint signals from individual ones, especially when the individual signals are closely aligned. Finally, even when the individual signals are orthogonal, the estimated product may still have small but non-zero singular values, with their magnitudes increasing as the noise level rises.

One of our main theoretical contributions lies in the precise characterization of perturbations in the spectrum of $\widehat{M} := P_{\mathrm{col}(\widehat{X}_1)} P_{\mathrm{col}(\widehat{X}_2)}$. Our result provides general guarantees that hold regardless of the estimation procedure used to approximate the signals $(\widehat{X}_1, \widehat{X}_2)$. The quality of the chosen estimation procedure is reflected in our analysis via deterministic quantities $\varepsilon_1 := \left\|P_{\mathrm{col}(X_1)}\left(\Delta_1 + \Delta_2 + \Delta_1\Delta_2\right) P_{\mathrm{col}(X_2)}\right\|_2$ and $\varepsilon_2 := \left\|P_{\mathrm{col}(X_1)}\Delta_2 + \Delta_1 P_{\mathrm{col}(X_2)} + \Delta_1\Delta_2\right\|_2$ with $\Delta_k := P_{\mathrm{col}(X_k)} - P_{\mathrm{col}(\widehat{X}_k)}$. The proof is in the Appendix A.1.

Theorem 1 (Spectrum of the product of projections). *Let* $\tau_{\min} := \sigma_{\min}(P_{\mathcal{N}_1} P_{\mathcal{N}_2})$ *and* $\tau_{\max} := \sigma_{\max}(P_{\mathcal{N}_1} P_{\mathcal{N}_2}) = \left\|P_{\mathcal{N}_1} P_{\mathcal{N}_2}\right\|_2$. *If* $\mathrm{rank}(X_1), \mathrm{rank}(X_2) > 0$*, then the spectrum of* $\widehat{M}$ *is:*

1. *a set of* $\dim(\mathcal{J})$ *singular values in* $[1 - \varepsilon_1 \vee 0, 1]$;
2. *a set of* $\mathrm{rank}\left(P_{\mathcal{N}_1} P_{\mathcal{N}_2}\right)$ *singular values in* $[(\tau_{\min} - \varepsilon_1) \vee 0, (\tau_{\max} + \varepsilon_2) \wedge 1]$;
3. *a remaining set of singular values in* $[0, \varepsilon_2 \wedge 1]$.

This theorem states that in the presence of a signal, the spectrum of the product of projection matrices $\widehat{M}$ organizes into three sets: joint, non-orthogonal individual, and noise directions. Here ε_1 quantifies the maximal downward perturbation of the singular values associated with the joint signal—i.e., how much true singular values of one may shrink toward zero due to noise. These perturbed singular values fall within $[1 - \varepsilon_1 \vee 0, 1]$. In contrast, ε_2 captures the maximal upward perturbation of null (noise-only) singular values, which may become spuriously large due to random overlap; these lie in $[0, \varepsilon_2 \vee 1]$. When these intervals are disjoint, joint and noise directions can be perfectly separated. More specifically, the constraint $\varepsilon_1 < \max\{1 - \varepsilon_2, 1 - \left\|P_{\mathcal{N}_1} P_{\mathcal{N}_2}\right\|_2 - \varepsilon_2\}$ ensures that the singular values corresponding to the joint structure are distinguishable from other clusters, that is the number of singular values above $1 - \varepsilon_1$ correctly identifies the joint rank. Similarly, the constraint $\varepsilon_1 + \varepsilon_2 < \sigma_{\min}(P_{\mathcal{N}_1} P_{\mathcal{N}_2})$ ensures that the set of singular values corresponding to the non-orthogonal individual structure is distinguishable from the noise set. When both constraints hold, the spectrum of $\widehat{M}$ has *three non-overlapping clusters* corresponding to joint, non-orthogonal individual, and noise directions. The gap between the clusters decreases with larger values of $\varepsilon_1, \varepsilon_2$ and larger alignment of the non-orthogonal individual subspaces $\mathcal{N}_1$ and $\mathcal{N}_2$. Figure 1 illustrates these theoretical intervals alongside observed singular values of $\widehat{M}$ and true singular values of $P_{\mathrm{col}(X_1)} P_{\mathrm{col}(X_2)}$.

The error terms ε_1, ε_2 depend on the noise level: higher noise leads to larger perturbations Δ_k in estimating each view, and thus larger values of $\varepsilon_1, \varepsilon_2$. Figure 1 shows how a smaller signal-to-noise ratio results in smaller cluster gaps. The estimated subspace dimension $\mathrm{col}(\widehat{X}_k)$ also affects the size of ε_1, ε_2. Underestimating the rank (e.g., $\mathrm{rank}(\widehat{X}_1) < \mathrm{rank}(X_1)$) can lead to large ε_1 and ε_2, especially if missing directions are aligned with both $\mathrm{col}(X_1)$ and $\mathrm{col}(X_2)$. In fact, $\varepsilon_1 = 1$ when a missing direction is part of the joint structure, in which cases the effective number of large singular values corresponding to the joint structure gets smaller than the true joint rank. Overestimating the rank (e.g., $\mathrm{rank}(\widehat{X}_1) > \mathrm{rank}(X_1)$) can also lead to large errors due to increased dimensionality of Δ_k, making the third component of ε_2, $\|\Delta_1\Delta_2\|_2$, very large, albeit slight overestimation would still keep those terms small. Thus, overestimation is generally preferred to underestimation. We use these insights in § 4, providing practical guidelines on marginal rank estimation.

The clustering of the singular values of $\widehat{M}$ enables a natural estimator for the signal subspaces. The joint subspace dimension is estimated as the number of singular values of $\widehat{M}$ above $1-\varepsilon_1$, denoted by $\widehat{r}_{\mathcal{J}}$. According to Theorem 1, this estimate is exact if $\varepsilon_1 < 1 - \|P_{\mathcal{N}_1}P_{\mathcal{N}_2}\|_2 - \varepsilon_2$. To estimate the joint subspace itself, a candidate approach is to use the span of the first $\widehat{r}_{\mathcal{J}}$ left (or right) singular vectors of $\widehat{M}$. However, this approach arbitrarily favors one view over the other, insisting that the joint subspace is strictly contained in $\mathrm{col}(\widehat{X}_1)$. To avoid this, we symmetrize the product of projection matrices by $\widehat{S} := \frac{1}{2}(\widehat{M} + \widehat{M}^\top)$, which effectively averages the estimated joint directions from each view, yielding a more accurate estimate. We estimate the joint and individual subspace structures as

$$\begin{aligned}
\widehat{\mathcal{J}} &:= \text{span of the first } \widehat{r}_{\mathcal{J}} \text{ singular vectors of } \widehat{S} = \frac{1}{2}\left(P_{\mathrm{col}(\widehat{X}_1)}P_{\mathrm{col}(\widehat{X}_2)} + P_{\mathrm{col}(\widehat{X}_2)}P_{\mathrm{col}(\widehat{X}_1)}\right), \\
\widehat{\mathcal{I}}_k &:= \text{span of the first } \mathrm{rank}(\widehat{X}_k) - \widehat{r}_{\mathcal{J}} \text{ left singular vectors of } P_{\mathrm{col}(\widehat{X}_k)}(I - P_{\widehat{\mathcal{J}}}).
\end{aligned} \tag{3}$$

We use the symmetrized $\widehat{S}$ to be invariant to the order of views with coinciding left and right singular vectors. We further characterize the quality of these estimates, the proof is in Appendix A.1.

THEOREM 2 (ESTIMATION ERROR OF THE JOINT AND INDIVIDUAL SUBSPACES). *Define* $R_{\mathcal{J}} := P_{\mathrm{col}(X_1)}\Delta_2 + \Delta_1 P_{\mathrm{col}(X_2)} + \Delta_1\Delta_2$ *and* $R_{\mathcal{I}_k} := P_{\mathcal{J}^\perp}\Delta_k - \Delta_{\mathcal{J}}P_{\mathrm{col}(X_k)} + \Delta_{\mathcal{J}}\Delta_k$, *where* $\Delta_{\mathcal{J}} := P_{\mathcal{J}} - P_{\widehat{\mathcal{J}}}$ *and* $\Delta_k := P_{\mathrm{col}(X_k)} - P_{\mathrm{col}(\widehat{X}_k)}$. *Suppose* $\varepsilon_1 < 1 - \|P_{\mathcal{N}_1}P_{\mathcal{N}_2}\|_2 - \varepsilon_2$. *Then,*

$$\dim(\widehat{\mathcal{J}}) = \dim(\mathcal{J}) \quad \textit{and} \quad \left\|P_{\mathcal{J}} - P_{\widehat{\mathcal{J}}}\right\|_2 \le \frac{\left\|R_{\mathcal{J}} + R_{\mathcal{J}}^\top\right\|_2}{1 - \|P_{\mathcal{N}_1}P_{\mathcal{N}_2}\|_2}.$$

Further, if the rank of each subspace view is correctly estimated, i.e. $\mathrm{rank}(\widehat{X}_k) = \mathrm{rank}(X_k)$,

$$\left\|P_{\mathcal{I}_k} - P_{\widehat{\mathcal{I}}_k}\right\|_2 \le 2\left\|R_{\mathcal{I}_k}\right\|_2.$$

This theorem quantifies the accuracy of estimated subspaces when the ranks are correctly specified. Since the spectral norm of the difference between two projection matrices equals the sine of the largest principle between the corresponding subspaces, our bounds measure the sine of the largest principle angles between $\mathcal{J}$ and $\widehat{\mathcal{J}}$, and between $\mathcal{I}_k$ and $\widehat{\mathcal{I}}_k$, respectively. For joint subspace estimation, the bound depends on the perturbation term $R_{\mathcal{J}}$ and the maximum alignment $\|P_{\mathcal{N}_1}P_{\mathcal{N}_2}\|_2$ between the individual subspaces. As in Theorem 1, large alignment between the individual subspaces can lead to significant errors. Note that the error due to such alignment also affects individual subspace estimation through the $\Delta_{\mathcal{J}}$ term in $R_{\mathcal{I}_k}$.

3.4. *Probabilistic bounds*

Theorems 1-2 are deterministic and hold for any signal estimate $\widehat{X}_k$ and for any data-generating noise Z_k in (1). In this section, we provide probabilistic bounds for a particular signal estimation procedure and noise model. Specifically, we let $\widehat{X}_k$ denote the rank-$\widehat{r}_k$ truncated singular value decomposition of Y_k, and consider Gaussian noise.

Assumption 1. (Gaussian noise) The entries of Z_k in (1) are independently and identically distributed as zero-mean Gaussian random variables with variance δ_k, $k = 1, 2$.

Assumption 2. (Signal-to-noise ratio) For each $k = 1, 2$, $\sigma_{r_k}(X_k) \gtrsim \delta_k(n^{1/2} + p_k^{1/2})$. That is, the smallest non-zero singular value of each view's signal is sufficiently large.

Under Assumptions 1 and 2, along with rank estimation conditions, we obtain the following probabilistic bounds in terms of the essential aspects of the problem (e.g., the sample size, the number of features in each view, signal ranks) and omit constants. The proof is provided in Appendix A.3.

Theorem 3 (Probabilistic bounds). *Suppose Assumptions 1-2 hold. Let* $r_{\max} := \max\{r_1, r_2\}$, $p_{\max} := \max\{p_1, p_2\}$, $p_{\min} := \min\{p_1, p_2\}$, $\delta_{\max} := \max\{\delta_1, \delta_2\}$, *and* $\sigma_r(X) := \min\{\sigma_{r_1}(X_1), \sigma_{r_2}(X_2)\}$. *If* $\widehat{r}_k \geq r_k$ *for* $k = 1, 2$, *then, with probability greater than* $1 - O(\exp\{-\max(n, p_{\min})\})$*:*

$$\epsilon_1 \lesssim \frac{\delta_{\max}(n^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)} + \frac{\delta_{\max}^2(n^{1/2} + p_{\max}^{1/2})(p_{\max}^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)^2}.$$

If $\widehat{r}_k = r_k$ *then with probability greater than* $1 - O(\exp\{-\max(n, p_{\min})\})$*:*

$$\epsilon_2 \lesssim \frac{\delta_{\max}(n^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)} + \frac{\delta_{\max}^2(n^{1/2} + p_{\max}^{1/2})(p_{\max}^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)^2}.$$

If additionally $\widehat{r}_J = r_J$, *then with probability greater than* $1 - O(\exp\{-\max(n, p_{\min})\})$*:*

$$\max\left\{\left\|\mathcal{P}_{\widehat{\mathcal{J}}} - \mathcal{P}_{\mathcal{J}}\right\|_2, \left\|\mathcal{P}_{\widehat{\mathcal{I}}_k} - \mathcal{P}_{\mathcal{I}_k}\right\|_2\right\} \lesssim \frac{1}{1 - \|P_{\mathcal{N}_1} P_{\mathcal{N}_2}\|_2}\left\{\frac{\delta_{\max}(n^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)} + \frac{\delta_{\max}^2(n^{1/2} + p_{\max}^{1/2})(p^{1/2} + r_{\max}^{1/2})}{\sigma_r(X)^2}\right\}.$$

Theorem 3 makes explicit the distinctive effect of rank estimation on ε_1 and ε_2. Although we obtained identical bounds, the bound for ε_1 holds true for all $\widehat{r}_k \geq r_k$, whereas the bound for ε_2 is only true when $\widehat{r}_k = r_k$. Intuitively, rank overestimation does not adversely affect ε_1, as the alignment of joint directions is already captured. In contrast, adding extra noise dimensions increases the chance of a random strong alignment, pushing ε_2 to be high.

Assumption 2 is a signal-to-noise ratio assumption that is common in the random matrix theory due to known concentration $\|Z_k\|_2 \asymp \delta_k(n^{1/2} + p_k^{1/2})$ (Vershynin, 2025), with high probability. This leads to the standard bound $\|\Delta_k\|_2 \lesssim \delta_k(n^{1/2} + p_k^{1/2})/\sigma_{r_k}(X_k)$ via Davis–Kahan theorem (Davis & Kahan, 1970). Our bounds on magnitude of ε_k, however, are tighter. Specifically, the first term does not depend on the number of features p_k, and the dependence on p_k is only in the second term, which is of smaller order when the signal $\sigma_{r_k}(X_k)$ is large. Instead, we have the explicit dependence on the rank r_k, which highlights how smaller-dimensional signals are easier to estimate.

The third bound highlights the difficulties that arise from the non-orthogonal individual subspaces $\mathcal{N}_k$. As expected, high alignments across these subspaces makes the separation of joint and individual structures more difficult, affecting estimation accuracy.

We note that both the error terms ε_1 and ε_2 go to zero in probability as long as the signal-to-noise ratio goes to infinity, that is $\sigma_r(X_k)/\{\delta_k(n^{1/2} + p_k^{1/2})\} \to \infty$ as $n \to \infty$ for $k = 1, 2$. Additionally, the error terms $\left\|\mathcal{P}_{\widehat{\mathcal{J}}} - \mathcal{P}_{\mathcal{J}}\right\|_2$ and $\left\|\mathcal{P}_{\widehat{\mathcal{I}}_k} - \mathcal{P}_{\mathcal{I}_k}\right\|_2$ go to zero under a similar condition.

4. Proposed Algorithm Based on Product of Projection Matrices

4.1. *Overview*

In practice, our primary goal is to use the observed spectrum of $P_{\mathrm{col}(\widehat{X}_1)}P_{\mathrm{col}(\widehat{X}_2)}$ to accurately determine the joint rank. Our new theoretical insights characterize spectrum perturbation in terms of ε_1 and ε_2, both of which are unknown in practice and depend on the estimates $\widehat{X}_k$. Specifically, Theorem 1 tells that singular values above $1-\varepsilon_1$ correspond to the joint, while those below ε_2 correspond to the noise directions. Therefore, we focus on the following practical questions: (i) how to estimate $\mathrm{col}(\widehat{X}_k)$ for the signal X_k in each view $k \in \{1,2\}$? (ii) how to estimate ε_1 in practice to identify the top cluster of potential joint directions? (iii) how to bound ε_2 to ensure the top cluster contains only joint directions and avoid overlap from random noise alignment? From the discussion in Section 3.3, we also have the following insights: (i) rank overestimation is preferred to underestimation since it leads to smaller ε_1; (ii) simultaneously, overestimation pushes ε_2 closer to 1 which makes it harder to distinguish the joint signal from the noise; (iii) rank-to-dimension ratio, signal-to-noise (snr) ratio, and individual component alignment all have influence on the final estimation quality – in general, high signal rank, low snr, and high degree of alignment make the problem highly intractable. Below, we outline answers to these practical questions and summarize our proposed method.

4.2. *Estimation of* $\mathrm{col}(X_k)$.

To estimate the signal X_k, we construct $\widehat{X}_k$ using truncated singular value decomposition of observed Y_k for each view $k \in \{1,2\}$, an approach which allowed us to derive probabilistic bounds in Section 3.4. Since overestimation is preferred over underestimation, we choose the level of truncation based on the criterion in Gavish & Donoho (2014), which tends to overestimate the signal rank. Let $\beta_k = n/p_k$ be the aspect ratio, and y_{med} be the median singular value of Y_k. Then we only keep those singular vectors whose corresponding singular values exceed the threshold $(0.56\beta_k^3 - 0.95\beta_k^2 + 1.82\beta_k + 1.43)\widehat{\sigma}_k$. This threshold is designed to obtain an optimal (in mean squared error) reconstruction of the signal matrix. Because severe overestimation is still problematic as it introduces potentially many noise directions, we show how to filter out these directions in part (iii) while also we provide a diagnostic plot that allows to confirm the choice.

Estimation of ε_1 via rotational bootstrap. From Theorem 1, the number of singular values of $\widehat{M} = P_{\mathrm{col}(\widehat{X}_1)}P_{\mathrm{col}(\widehat{X}_2)}$ exceeding $1-\varepsilon_1$ provides an estimate for the joint rank, $r_{\mathcal{J}}$. This approach, however, requires an estimate for the threshold parameter ε_1. To this end, we propose a novel rotational bootstrap procedure. Previous bootstrap procedures, e.g., in Prothero et al. (2024), independently resample each view's signal subspace, U_k. This fails to preserve the alignment between U_1 and U_2, since in high-dimensional settings, independently sampled subspaces are likely to be nearly orthogonal, particularly when the rank-to-dimension ratio is small. As a result, independent bootstraping leads to underestimation of ε_1, and consequently, an unreliable estimate of the joint rank.

In contrast, our proposed method circumvents this issue by sampling the subspaces jointly, thereby preserving their relative orientation. Specifically, let $\Sigma_{\widehat{M}}$ be a diagonal matrix containing singular values of $\widehat{M} = P_{\mathrm{col}(\widehat{X}_1)}P_{\mathrm{col}(\widehat{X}_2)}$. We take $[U_{1,(b)}, U_{2,(b)}]$ to be the first $\widehat{r}_1 + \widehat{r}_2$ left singular vectors of a Gaussian matrix N of size $n \times n$. We then align the signal subspaces by updating $U_{2,(b)}$ to be $U_{1,(b)}\cos(\Sigma_{\widehat{M}}) + U_{2,(b)}\sin(\Sigma_{\widehat{M}})$. By construction, the singular values of $U_{1,(b)}^\top U_{2,(b)}$ are exactly $\cos(\Sigma_{\widehat{M}})$. For each bootstrap sample, the data replicate is generated as $Y_{k,(b)} := U_{k,(b)}\widehat{\Sigma}_k V_{k,(b)}^\top + \widehat{E}_k$, where $\widehat{\Sigma}_k$ is a matrix of singular values from $\widehat{X}_k$, $V_{k,(b)} \in \mathbb{R}^{p\times\widehat{r}_k}$ are orthonormal, drawn uniformly and independently from the Haar measure, and $\widehat{E}_k$ is the adjusted estimate of the noise matrix (see Appendix C in Prothero et al. (2024)). Here $X_{k,(b)} = U_{k,(b)}\widehat{\Sigma}_k V_{k,(b)}^\top$ represents the signal replicate. Using data replicate $Y_{k,(b)}$, the corresponding truncated singular value decomposition of rank $\widehat{r}_k$ is given by $\widehat{X}_{k,(b)} := \widehat{U}_{k,(b)}\widehat{D}_{(b)}\widehat{V}_{k,(b)}^\top$. The resulting projection matrices $P_{\mathrm{col}(X_k),(b)} := U_{k,(b)}U_{k,(b)}^\top$ and $P_{\mathrm{col}(\widehat{X}_k),(b)} := \widehat{U}_{k,(b)}\widehat{U}_{k,(b)}^\top$ represent one bootstrap sample of the true and estimated signal subspaces, respectively, allowing estimation of principal angles over replications. Given B bootstrap replicates, we estimate ε_1 as:

$$\widehat{\varepsilon}_1 := \tfrac{1}{B}\textstyle\sum_{b=1}^{B} \|P_{\mathrm{col}(X_1),(b)}(\Delta_{1,(b)} + \Delta_{2,(b)} + \Delta_{1,(b)}\Delta_{2,(b)})P_{\mathrm{col}(X_2),(b)}\|_2, \tag{4}$$

where $\Delta_{k,(b)} = P_{\mathrm{col}(X_k),(b)} - P_{\mathrm{col}(\widehat{X}_k),(b)}$, $k \in \{1,2\}$.

4.3. *Filtering out noise directions via the random noise bound*

When the estimated subspaces $\mathrm{col}(\widehat{X}_1)$ and $\mathrm{col}(\widehat{X}_2)$ contain noise directions—a consequence of low signal-to-noise ratio or rank over-specification—these directions can exhibit spurious alignment purely by chance. Such random alignments produce a cluster of singular values bounded from above by ε_2, as established in Theorem 1. A large value of ε_2 complicates the separation of joint and noise subspaces, as it implies a potential overlap in their corresponding singular value distributions.

To filter out noise directions, one could adapt the bootstrap methodology to estimate ε_2 and use it as a filtering threshold. However, we find a simpler and more direct approach to be effective. By leveraging results from random matrix theory, we found a deterministic upper bound for ε_2 that depends only on the rank-to-dimension ratios.

To motivate the analytical bound, consider a simplified scenario with no individual signals, $P_{\mathrm{col}(X_K)} = P_{\mathcal{J}}$, with $P_{\mathrm{col}(\widehat{X}_k)} = P_{\mathcal{J}} + P_{\mathrm{noise}_k}$, that is the estimated signal is exactly the joint signal plus orthogonal noise directions. Formally, P_{noise_k} is the projection onto the subspace $\mathcal{J}^\perp \cap \mathrm{col}(\widehat{X}_k)$. Then $\varepsilon_2 = \left\|P_{\mathrm{col}(X_1)}\Delta_2 + \Delta_1 P_{\mathrm{col}(X_2)} + \Delta_1\Delta_2\right\|_2 = \left\|P_{\mathrm{noise}_1}P_{\mathrm{noise}_2}\right\|_2$, representing the maximal singular value of the product of random projections. While ε_2 is generally more complex, the key intuition remains: only singular values above the threshold expected from random noise alignment can be confidently identified as the signal.

We explicitly characterize the distribution of the entire spectrum of the product of random projections $P_{\mathrm{noise}_1}P_{\mathrm{noise}_2}$ by leveraging results from random matrix theory, which leads to the proposed analytical bound on $\left\|P_{\mathrm{noise}_1}P_{\mathrm{noise}_2}\right\|_2$. Let P_{noise_1} and P_{noise_2} denote two independent projection matrices onto random subspaces (drawn from a Haar measure) of $\mathbb{R}^n$ with ranks r_1 and r_2. Let $q_k := r_k/n$ for $k \in \{1,2\}$ be the aspect ratio. Then the asymptotic distribution of λ, the squares of the singular values of $P_{\mathrm{noise}_1}P_{\mathrm{noise}_2}$, as $n \to \infty$ (Bouchaud et al., 2007) has a continuous part

$$f(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\lambda(1-\lambda)}, \quad \lambda_\pm = q_1 + q_2 - 2q_1q_2 \pm 2\sqrt{q_1q_2(1-q_1)(1-q_2)}, \tag{5}$$

and two delta peaks $A_0\delta(\lambda)$ and $A_1\delta(\lambda - 1)$ with $A_0 = 1 - q_1 \wedge q_2$ and $A_1 = (q_1 + q_2 - 1) \vee 0$. We illustrate this distribution in Appendix B.4 as a function of q_1, q_2. As the number of noisy directions increases, so does the maximal singular value $\left\|P_{\mathrm{noise}_1}P_{\mathrm{noise}_2}\right\|_2$, reaching the value of one at $q_1 = q_2 = 1/2$. When $q_1, q_2 < 1/2$, the maximal singular value is given by $\lambda_+^{1/2}$ in (5).

This analytical result provides a practical filtering strategy. We propose to classify only those singular values as joint signal whose magnitude exceeds the spectral edge λ_+ (in addition to exceeding $1 - \widehat{\varepsilon}_1$). In practice, we calculate the aspect ratios q_k using the full estimated marginal ranks, $\mathrm{rank}(\widehat{X}_k)$. This substitution makes the bound conservative, as it provides a worst-case noise estimate by assuming all non-joint directions could be noise.

4.4. *Full algorithm description and diagnostics*

In summary, given the observed data matrices (Y_1, Y_2), we obtain estimates of $\mathrm{col}(\widehat{X}_1)$ and $\mathrm{col}(\widehat{X}_2)$ as in (i) to construct $\widehat{M} = P_{\mathrm{col}(\widehat{X}_1)} P_{\mathrm{col}(\widehat{X}_2)}$. We estimate the rank of the joint subspace $\widehat{r}_{\mathcal{J}}$ as the number of singular values in the spectrum of $\widehat{M}$ that are above $1 - \widehat{\varepsilon}_1 \vee \lambda_+^{1/2}$ where $\widehat{\varepsilon}_1$ and λ_+ are given by (4) and (5), respectively. We estimate the joint and individual subspaces according to (3). We summarize the steps in Appendix A.4.

We accompany our algorithm with a diagnostic plot (e.g., Figure 2 for COAD dataset), which illustrates the spectrum of the product $\widehat{M}$ together with the bounds $1 - \widehat{\varepsilon}_1$ and $\lambda_+^{1/2}$. The best case scenario is observing three clear clusters in the spectrum (see e.g., Figure 1 case of Angle = 50, SNR = 22), which provides distinct separation of joint, non-orthogonal individual, and noise irrespective of threshold estimation. More commonly, we expect to only see two clusters (noise and individual/joint). If the cluster of top values is strictly above the threshold, we can be confident those are joint. In case the cluster is in the middle of the threshold, it likely contains a combination of joint and individual. A lack of clustering structure would imply a large magnitude of noise, preventing accurate separation. In the latter case, driven by our understanding of the magnitude of $\|P_{\mathrm{noise}_1} P_{\mathrm{noise}_2}\|_2$ from Appendix B.4, we suggest further decreasing the initial marginal rank estimates to reduce aspect ratios q_1, q_2, which subsequently reduce λ_+. In § 6, we illustrate these ideas when analyzing real data.

5. Simulations on Synthetic Data

We use simulated data to evaluate the performance of our proposed product of projections decomposition method for joint and individual structure identification. For comparison, we consider methods of Lock et al. (2013), Feng et al. (2018), Gaynanova & Li (2019), Shu et al. (2019) and Park & Lock (2020).

We simulate data from the model (1) with $Y_k = J_k + I_k + Z_k$, $k = 1, 2$, and $X_k = J_k + I_k$, all representing $n \times p_k$ matrices. We let $n = 50$, $p_1 = 80$, $p_2 = 100$, $\mathrm{rank}(J_k) = 4$, $\mathrm{rank}(I_1) = 5$, and $\mathrm{rank}(I_2) = 4$. We sample the singular values of J_k and I_k independently and identically from the uniform distribution. We sample the row space of J_k and I_k uniformly at random according to the Haar measure. To obtain the column spaces of J_k and I_k, we first generate an $n \times n$ matrix N where each entry is sampled independently from a normal distribution and let UDV^T be the corresponding singular value decomposition; we then set the column space of J_k to be $\mathrm{col}(U_{[:,1:4]})$ and the column space of I_1 to be $\mathrm{col}(U_{[:,5:9]})$; we then generate the column space of I_2 to be $\mathrm{col}(\cos(\phi) U_{[:,5:9]} + \sin(\phi) U_{[:,10:14]})$, where ϕ encodes the largest principle angle between the individual subspaces $\mathrm{col}(I_1)$ and $\mathrm{col}(I_2)$. We sample the entries of the noise matrix $Z_k \in \mathbb{R}^{n \times p_k}$ independently and identically from the Gaussian distribution $\mathcal{N}(0, s_k^2)$. From Theorem 5.32 of Vershynin (2010), the largest singular value of an $n \times p_k$ matrix whose entries are sampled from $\mathcal{N}(0, s^2)$ is bounded by $s(\sqrt{n} + \sqrt{p_k})$. Therefore, to control the signal-to-noise ratio (SNR), we take the noise variance for each view to be $s_k = \|X_k\|_2 / (\mathrm{SNR}(\sqrt{n} + \sqrt{p_k}))$.

The theoretical results of § 3 emphasize three main parameters affecting accurate identification of the subspaces: maximum principle angle between the individual subspaces ϕ, noise level, and the marginal ranks of estimated subspaces. To test the magnitude of each of these effects, we consider $\phi \in \{30°, 90°\}$ and $\mathrm{SNR} \in \{0.5, 2\}$. For the rank estimation, we consider three scenarios: (a) we let each method estimate the ranks based on the default method's implementation; (b) we supply marginal ranks that are smaller than the truth (under-specified rank); (c) we supply marginal ranks that are larger than the truth (over-specified rank). For (b) and (c), the miss-specification

Table 1: Mean F-scores (6) on simulated data, larger values correspond to more accurate estimation of joint and individual subspaces.

	Under-specified rank				Estimated rank				Over-specified rank			
	SNR = 2		SNR=0.5		SNR = 2		SNR=0.5		SNR = 2		SNR=0.5	
	90°	30°	90°	30°	90°	30°	90°	30°	90°	30°	90°	30°
JIV	–	–	–	–	8.27	7.88	4.74	4.33	–	–	–	–
AJI	8.54	7.65	5.07	4.56	9.81	8.55	3.85	3.67	9.17	7.70	5.01	4.70
SLI*	–	–	–	–	9.61	7.84	1.49	1.57	–	–	–	–
DCC	8.61	7.59	5.10	4.52	9.80	8.28	3.78	3.60	9.14	7.53	4.93	4.46
UNI*	–	–	–	–	6.73	6.05	4.34	4.05	–	–	–	–
PPD	8.44	7.86	5.12	4.61	9.91	9.75	3.86	3.67	9.19	8.01	5.07	4.67

All values are multiplied by 10. The results are averaged across 50 iterations. The maximum standard error of the mean is 0.03. We denote by (*) methods that do not allow supplying marginal ranks for each view. We denote Lock et al. (2013) as JIV, Feng et al. (2018) as AJI, Gaynanova & Li (2019) as SLI, Shu et al. (2019) as DCC, Park & Lock (2020) as UNI, and our method as PPD.

is achieved by sampling uniformly from $\{1, 2, 3\}$ and subtracting it from $\mathrm{rank}(X_k)$ in the under-specified setting or adding it to $\mathrm{rank}(X_k)$ in the over-specified setting, respectively. Since the implementation of Lock et al. (2013), Gaynanova & Li (2019) and Park & Lock (2020) does not allow specification of marginal ranks, we only implement (b) and (c) with methods of Feng et al. (2018), Shu et al. (2019) and our approach.

Choosing the right accuracy measure for unsupervised multi-view decomposition is an open challenge, as discussed in Gaynanova & Sergazinov (2024). Many approaches rely on the percent of variance explained or the marginal view reconstruction. However, the percent of variance explained is maximized by a separate singular value decomposition for each view, favoring a lack of joint structure and larger marginal ranks. On the other hand, signal reconstruction error, though unbiased, does not account for structural assumptions or misclassification of the joint directions as individual signals. We propose a new accuracy measure using the subspace version of the false and true discovery metric from Taeb et al. (2020). For an estimated subspace $\widehat{\mathcal{S}}$ of a true subspace $\mathcal{S}$, we evaluate the true positive proportion (TPP) and false discovery proportion (FDP). These are then combined into a single metric, the F-score:

$$\mathrm{TPP} = \frac{\mathrm{tr}(P_{\widehat{\mathcal{S}}} P_{\mathcal{S}})}{\dim(\mathcal{S})}, \quad \mathrm{FDP} = \frac{\mathrm{tr}\left((I - P_{\mathcal{S}})\, P_{\widehat{\mathcal{S}}}\right)}{\dim(\widehat{\mathcal{S}})}, \quad \text{F-score} = 2 \cdot \frac{(1 - \mathrm{FDP}) \cdot \mathrm{TPP}}{1 - \mathrm{FDP} + \mathrm{TPP}}. \tag{6}$$

We compute the F-scores for joint and individual subspace estimates and take their average F-score as a final accuracy measure, with larger values indicating better performance.

Table 1 displays mean F-score values across 50 replications for each combination of angle, signal-to-noise ratio, and rank specification. The results align with what is expected based on Theorem 1: for each method, the performance deteriorates with larger alignment between individual subspaces (smaller angle), and smaller signal-to-noise ratio. In the large signal-to-noise ratio case, rank under-specification is worse than rank over-specification, which is also in agreement with the theory. In the low signal-to-noise ratio case, the results for default rank estimation are worse than those with controlled rank miss-specification, which we suspect is due to significant challenges in estimating rank correctly in this setting, with default estimation likely leading to even larger rank estimation error than controlled miss-specification. In terms of comparison across methods, it is difficult to imagine a uniformly best-performing one since different approaches have different strengths under different scenarios. However, our proposed method achieves the highest accuracy in most settings, followed closely by Feng et al. (2018)'s method. A more detailed empirical and theoretical comparison with Feng et al. (2018) is in Appendix A.6, where we show that our method can more effectively distinguish between joint and individual signals.

6. REAL DATA EXPERIMENTS

6.1. *Colorectal cancer data*

We analyze colorectal cancer data from the Cancer Genome Atlas with two distinct data views: RNA sequencing (RNAseq) normalized counts and miRNA expression profiles (Guinney et al., 2015). We process the data as in Zhang & Gaynanova (2022), leading to $p_1 = 1572$ RNAseq and $p_2 = 375$ miRNA variables. Colorectal cancer has been classified into four consensus molecular subtypes based on gene expression data (Guinney et al., 2015), and we obtain the labels from the Synapse platform (Synapse ID syn2623706). After filtering for complete data, we arrive at $n = 167$ samples from two views with cancer subtype information. Our goal is to extract joint and individual signals and determine which signals are most predictive for cancer subtypes. We apply the proposed product of projections methods and compare it with other methods from § 5. We also consider standard principal component analysis for each view as the baseline.

First, we determine the ranks corresponding to joint and individual structures for each method. For the proposed approach, we estimate the marginal ranks using the truncation method of Gavish & Donoho (2014). The scree plot for each view is shown in Figure 2 (left), resulting in ranks of 40 and 42, respectively. We then apply the proposed method, generating the diagnostic plot of the observed product of the projections spectrum (Figure 2, middle). In the plot, the blue region represents the noise cutoff $\lambda_+^{1/2}$, and the green region marks the top cluster based on the bootstrap estimate of ε_1. Observe that the spectrum does not show clear clustering, with the noise directions falling well within the region identified by the bootstrap. This inconsistency is likely due to the high rank-to-dimension ratios, which inflate both bounds and the low signal-to-noise ratio, which affects the bootstrap estimate. To address the inclusion of noise directions, we re-examined the scree plot presented in Figure 2. The initial marginal ranks appeared too high, so we identified the elbow point at rank 16 for each view based on the scree plot.

After adjusting the ranks, we re-applied the proposed approach with the updated diagnostic plot (Figure 2, right). The updated plot shows more distinct clustering, with a clear separation between lower and higher singular values. Both the bootstrap and noise bounds support this observation by identifying the cluster of high singular values as the joint structure. This analysis highlights the effectiveness of the proposed diagnostic plot, enabling practitioners to fine-tune parameters to obtain a more confident separation of joint directions.

Next, we evaluate the predictive power of the estimated joint and individual components for cancer subtype classification using multinominal regression. For a fair comparison, we use the same marginal ranks as ours for Feng et al. (2018), Shu et al. (2019), and principal component analysis. For other methods, we use their default rank estimation procedures, as custom ranks could not be specified. We assess classification performance separately for the joint and the individual subspaces. We expect that the joint components will be closely related to the cancer subtype, while the individual components would capture differences between RNA and mRNA platforms unrelated to subtypes. Table 2 shows the estimated ranks, miss-classification error rates, and the smallest angle between individual subspaces for each method. As expected, most methods perform better using joint components than individual ones. The exceptions are principal components analysis and the methods of Lock et al. (2013) and Park & Lock (2020). For principle components analysis and Lock et al. (2013), the angles between individual subspaces are small,

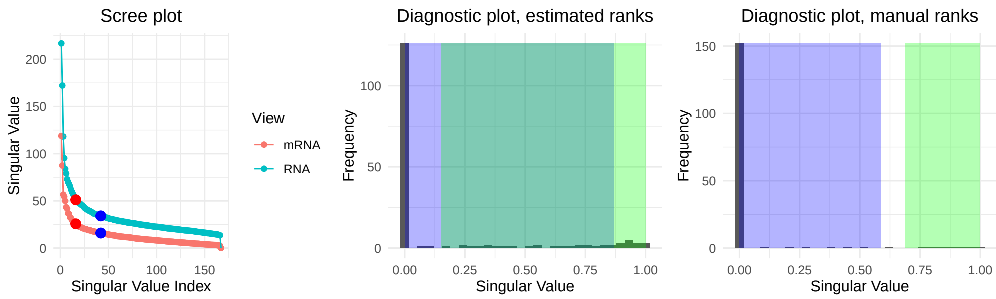

Fig. 2: Left: Scree plot for RNASeq and mRNAseq data. Blue point denotes the cutoff suggested by Gavish & Donoho (2014). Red point indicates our suggested cutoff. Middle / Right: Spectrum of the product of projection matrices onto each view for two different marginal ranks. The blue region indicates the random direction bound, $[0, \lambda_+^{1/2}]$, used to filter out the noise directions. The green region is given by $[1-\widehat{\varepsilon}_1, 1]$ where $\widehat{\varepsilon}_1$ is a bootstrap estimate of ε_1.

Table 2: Mean cancer subtype misclassification rates in percentages together with the minimum principle angle between estimated individual subspaces.

	Joint ($\mathcal{J}$)		RNAseq ($\mathcal{I}_1$)		miRNA ($\mathcal{I}_2$)		Angle
	Rank	Error	Rank	Error	Rank	Error	$\angle(\mathcal{I}_1, \mathcal{I}_2)$
PCA	–	–	16	0	16	6.6	2
JIV	2	44.3	20	0	13	30.5	16
AJI	9	7.2	7	50.9	8	61.7	57
SLI	32	0	41	43.7	10	59.9	90
DCC	9	7.8	7	53.3	7	66.5	90
UNI	1	55.1	5	0	1	44.3	74
PPD	8	8.9	8	53.9	8	62.9	52

We denote principle components analysis by PCA, Lock et al. (2013) as JIV, Feng et al. (2018) as AJI, Gaynanova & Li (2019) as SLI, Shu et al. (2019) as DCC, Park & Lock (2020) as UNI, and our method as PPD.

suggesting that the joint structure may be mistaken for the individual. For Park & Lock (2020), the estimated marginal rank of miRNA view is much smaller compared to all other approaches, so it is possible that the joint structure was missed in that view. For joint subspace, Gaynanova & Li (2019) achieves the lowest misclassification rate; however, the corresponding estimated ranks are much higher than for other approaches. In comparison, our method offers a more parsimonious representation of the joint components while retaining strong predictive power, effectively balancing simplicity and informativeness. Feng et al. (2018) and Shu et al. (2019) achieve slightly better miss-classification rates (by 1%) but at the cost of including an additional joint component. From the angles between the individual subspaces reported in Table 2, we see that this additional component corresponds to aligned individual directions with an angle of 52 degrees between the views, which is excluded from the joint subspace in our approach. Indeed, in Figure 2, this angle corresponds to the rotated individual directions by falling exactly in-between our noise bound, $\lambda_+^{1/2}$, and bootstrap bound, $1-\varepsilon_1$. This again highlights the value of the diagnostic plot and underscores the clear subspace separation achieved by our method.

6.2. *Nutrigenomic mice data*

The nutrigenomic mice data (Martin et al., 2007) contain measurements from $n = 40$ mice from two views: gene expressions ($p_1 = 120$) and lipid content of the liver ($p_2 = 21$). The data

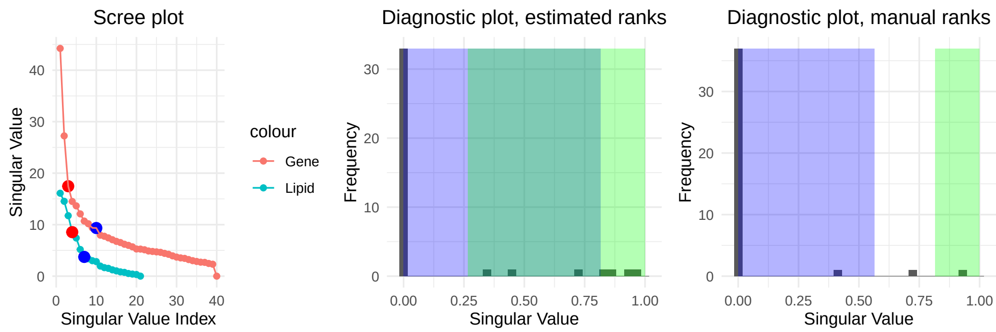

Fig. 3: Left: Scree plot for gene expression and lipid content data. Blue point denotes the cutoff suggested by Gavish & Donoho (2014). Red point indicates our suggested cutoff. Middle / Right: Spectrum of the product of projection matrices onto each view for two different marginal ranks. The blue region indicates the random direction bound, $[0, \lambda_+^{1/2}]$, used to filter out the noise directions. The green region is given by $[1 - \widehat{\varepsilon}_1, 1]$ where $\widehat{\varepsilon}_1$ is a bootstrap estimate of ε_1.

Table 3: Mean genotype and diet misclassification rates in percentages together with the minimum principle angle between estimated individual subspaces.

	Joint ($\mathcal{J}$)			Gene ($\mathcal{I}_1$)			Lipid ($\mathcal{I}_2$)			Angles
	Rank	Type	Diet	Rank	Type	Diet	Rank	Type	Diet	$\angle(\mathcal{I}_1, \mathcal{I}_2)$
PCA	–	–	–	3	0	60	4	0	0	22
JIV	2	0	27.5	3	25	57.5	3	30	12.5	44
AJI	2	0	55	1	52.5	77.5	2	47.5	10	66
SLI	6	0	0	8	45	42.5	1	50	57.5	90
DCC	3	0	37.5	1	45	65	1	45	47.5	85
UNI	2	0	62.5	1	52.5	77.5	2	35	0	72
PPD	1	0	80	2	37.5	57.5	3	30	0	44

We denote principle components analysis by PCA, Lock et al. (2013) as JIV, Feng et al. (2018) as AJI, Gaynanova & Li (2019) as SLI, Shu et al. (2019) as DCC, Park & Lock (2020) as UNI, and our method as PPD.

are available as part of the multi-omics R package (Rohart et al., 2017). We process the data as in Yuan et al. (2022). The mice were selected from two genotypes (wild-type and PPAR-α mutant). They were administered five distinct diets consisting of corn and colza oils (ref), a saturated fatty acid diet of hydrogenated coconut oil (coc), an Omega6-rich diet of sunflower oil (sun), an Omega3-rich diet of linseed oil (lin), and a diet with enriched fish oils (fish). Our goal is to identify the joint and individual components across views. We expect that the joint components would be predictive of the mice genotype, while individual components for the lipid view would be predictive of the diet.

As in § 6.1, we start by determining the ranks for each view. The truncation method of Gavish & Donoho (2014) leads to ranks 11 and 7 for gene expressions and lipid concentrations, respectively. Figure 3, left, shows the corresponding scree plot, which suggests that the selected ranks may be too large. Our proposed diagnostic plot based on these ranks is displayed in Figure 3, middle. The noise bound is quite large, and overlaps with the bootstrap bound, suggesting that the rank reduction would be beneficial in filtering the noise directions. Previous analysis of these data has used ranks 3 for genes and 4 for lipids (Yuan et al., 2022). These ranks appear to better align with an elbow on the scree plot and lead to a diagnostic plot with a clear clustering structure and lack of overlap (Figure 3, right). We thus use these ranks for subsequent analysis.

We evaluate the predictive power of the estimated joint and individual components separately for the classification of genotype and diet based on multinomial linear regression. Table 3 shows that the identified joint components for all methods perfectly separate the genotype, with our method providing the most parsimonious representation of rank equal to one. Predicting the diet from the individual lipid features turns out to be a harder task, with only our method and that of Park & Lock (2020) achieving the perfect classification. While principle components analysis also achieves perfect diet classification, the angle between individual components of 22 degrees is small and suggests a missing joint structure. A closer look at Figure 3 shows that the perfect classification of diet by our method is based on the rotated individual direction, which is not part of our joint estimate. In contrast, Feng et al. (2018)'s method treats this direction as a joint component, leading to worse diet classification accuracy. As in the example from Section 6.1, the diagnostic plot proves valuable in evaluating the potential over-specification of ranks and providing confidence in the separation of joint and individual components.

7. Conclusion

In this work, we describe a multi-view data decomposition method that naturally derives from the identifiability conditions for joint and individual subspaces. The advantages of our approach include subspace estimation guarantees, scalability, and interpretable diagnostics.

Our work opens several avenues for future research. First, it would be valuable to rigorously extend our method to the multi-view setting with partially shared joint signals. Our current extension in Appendix A.5 assumes fully shared joint components and selects the joint rank via pairwise testing. Second, further theoretical developments could elucidate the properties of our rotational bootstrap approach and refine the asymptotic characterization of the noise bound to account for non-random subspaces. Finally, since many practical applications emphasize predictive performance, extending our method to the semi-supervised setting, as in Park et al. (2021); Palzer et al. (2022), would be beneficial.

8. Acknowledgements

Armeen Taeb is supported by NSF DMS-2413074 and by the Royalty Research Fund at the University of Washington. Irina Gaynanova's research was supported by NSF DMS-2422478.

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